

Samita

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Professional Summary

Computer Science graduate (Data Science) with hands-on experience across machine learning / computer vision and full-stack software engineering. Built production mobile apps (React Native/Expo, native Kotlin modules), AI-driven features (OpenAI APIs, YOLO-based CV), and backend/admin systems (Node.js, MongoDB, Next.js). Holds a patent for an ML-based posture monitoring system.

Professional Experience

Samsara Wellness

Dec 2025 – June 2026

ML Engineer / Software Development Engineer

Bengaluru, India

Admin Dashboard – Membership & Health CRM System (MERN)

- Built an admin dashboard using Next.js (TypeScript) to manage users, memberships, and health data at scale.
- Developed membership APIs in Node.js + Express + MongoDB for trial plans, lifetime plans, coupon-based access, renewals, cancellations, and expiry tracking.
- Created a separate admin service layer to fetch per-user active memberships and history, integrating it into a 100+ user listing table.
- Implemented period tracking & prediction features, including cycle history, phase detection, and upcoming period forecasts.
- Integrated secure authenticated API calls, handled async parallel data fetching, and resolved route mismatches, ES module export issues, 404 errors, and frontend-backend field-mapping bugs.
- Debugged real-world production issues involving API responses, schema mismatches, date handling, and multi-user state synchronization.

Health & Wellness Mobile Apps – HealthSync, PulseOnly, PulseApp (React Native, Kotlin, Expo)

- **HealthSync:** Built a cross-platform app aggregating steps, heart rate, calories, and sleep data from Android Health Connect and Apple HealthKit into a single dashboard; developed a custom native Kotlin module (via React Native's New Architecture) to bridge Health Connect on Android, and used Nitro Modules to query HealthKit on iOS (including Apple Watch data); configured Expo config plugins and EAS Build profiles for cross-platform CI/CD.
- **PulseOnly:** Built a camera-based heart-rate (PPG) monitor using a custom native Kotlin module (CameraX + Camera2 interop) to stream and analyze camera frames; implemented a real-time signal-processing pipeline (YUV→RGB conversion, Butterworth bandpass filtering, autocorrelation-based peak detection) to compute BPM, bridged to a live-scanning React Native UI via NativeEventEmitter.
- **PulseApp:** Built a multi-feature wellness app combining the pulse scanner, Ayurveda chatbot, and barcode/food scanners in a single tab-navigation flow, with image capture/compression, AsyncStorage-based local caching with automatic expiry, and a custom-themed UI.

Samsara Wellness

Sept 2025 – Nov 2025

ML and SDE Intern

Bengaluru, India

AI-Powered App Features – Ayurveda Bot, Barcode Scanner, Food Scanner

- Developed an Ayurveda Bot: a domain-specific chatbot built on the OpenAI Assistants API, trained on curated Ayurveda knowledge for personalized wellness guidance; designed the chat UI, imple-

mented conversation threads, and optimized API usage with streaming responses.

- Implemented an AI-driven Barcode Scanner that analyzes scanned products and returns product details, allergens, vegetarian classification, harmful-ingredient analysis, and structured nutrition tables.
- Built a Food Scanner module that detects multiple foods in a single image and returns per-gram macros (carbs, fats, protein, fiber), with total nutrition summaries and a clean card-based UI.
- Worked extensively with React Native + Expo, resolving image-upload issues, FormData bugs, Android-specific path/permission errors, deprecated ImagePicker APIs, "Network request failed" errors, forced dark-mode theming in production, and incorrect App Ownership in Expo; produced stable production APKs via Expo EAS.

Inonetecx Inc.

Dec 2024 – Mar 2025 (Remote)

Machine Learning & Web Development Intern

Toronto, Canada

- Developed and maintained a Flask-based web application, contributing to backend logic, feature workflows, and modular service design.
- Implemented a complete user wishlist system, allowing users to create wishlists, add products, and manage saved items with seamless UI integration.
- Built an automated service that fetches and extracts product details from shared product links using Selenium for text data scraping, reducing manual input and improving user convenience.

Projects

ML/DL-Based Yoga Pose Recognition System

Patented

YOLOv10, MediaPipe, OpenCV, Roboflow

- Developed a real-time yoga pose recognition system using YOLOv10 Nano, OpenCV, and 17 human body keypoints (shoulders, hips, elbows, knees, ankles, etc.) to analyze posture alignment.
- Collected, annotated, and labeled a custom dataset of 2,500–3,000 images using Roboflow, including preprocessing, augmentation, and generating YOLO-compatible keypoint annotations.
- Achieved 92% accuracy, 88% precision, and 91% recall on the curated keypoint dataset.
- Awarded Indian Patent No. 202511067136A for the posture monitoring and alignment-correction approach.

Loan Recommendation System

Python, Scikit-learn

- Built a predictive ML model to assess loan eligibility based on credit score, applying data preprocessing, feature engineering, and visualization techniques.

Education

Quantum University, Roorkee

2022 – 2026

B.Tech in Computer Science & Engineering (Data Science)

CGPA: 8.13/10

Technical Skills

- **Languages:** Python, JavaScript/TypeScript, SQL, C++, HTML, CSS
- **ML / Data Science:** TensorFlow, PyTorch, OpenCV, YOLOv10, NumPy, Pandas, Data Annotation, Matplotlib, Seaborn, Power BI
- **Web & App Development:** React.js, React Native (Expo), Node.js, Express.js, Next.js, Flask, Spring Boot
- **Databases:** MongoDB, PostgreSQL, MySQL
- **Mobile / Native:** Kotlin, CameraX, Android Health Connect, Apple HealthKit

- **Tools & Platforms:** Git, GitHub, VS Code, Jupyter Notebook, Linux, EAS Build (CI/CD)

Certifications & Achievements

- The Complete Python Bootcamp – Python 3, OOP & Flask (Udemy)
- Machine Learning A-Z: AI, Python & Data Science (Udemy)
- Python and Flask Bootcamp – REST APIs with Flask (Udemy)
- Patented Yoga Posture Monitoring System (Patent No. 202511067136A)